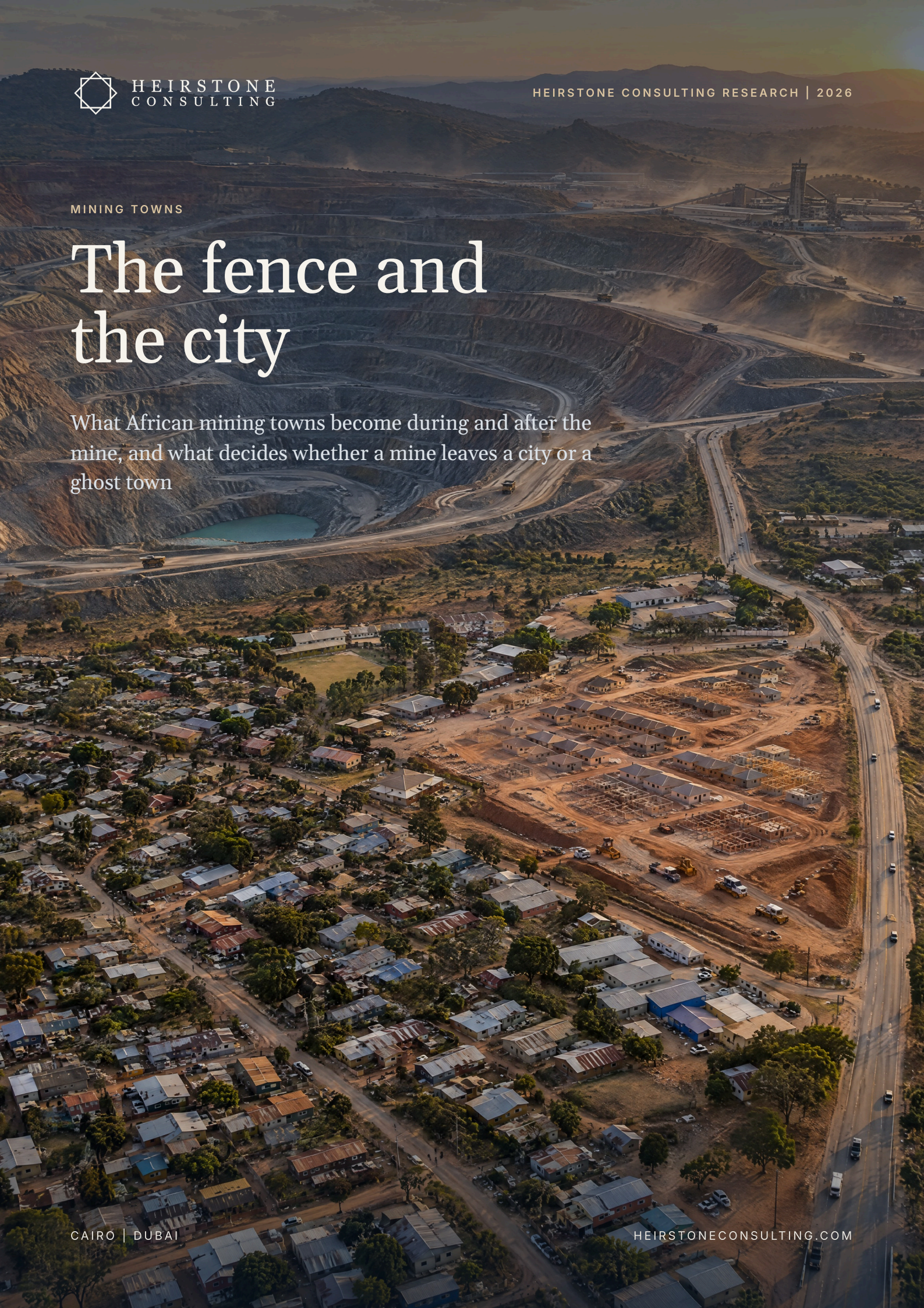
HEIRSTONE CONSULTING

MINING TOWNS

The fence and the city

What African mining towns become during and after the mine, and what decides whether a mine leaves a city or a ghost town



SIX NUMBERS FROM THIS REPORT

8 of 25

mining towns in the Heirstone panel that grew faster than their country's urban population over the same census years. Six shrank.

6 of 6

towns in the panel that lost residents were company towns, built by or for a single operator (Heirstone classification)

7.9x

growth of Solwezi, host town of the Kansanshi copper mine, between the 2000 and 2022 censuses (38,121 to 301,370; the 2022 count is compiled from ward data)

37%

of the legal royalty share due in Katanga reached it in 2014. Ghana's mining fund received about 38% of its due share that year, on cash figures that may mix years.

4 to 1

Simandou's construction peak of over 60,000 workers against fewer than 15,000 needed to run its mines, railway and port, as reported by Reuters

87%

planned population of OCP's Green City at Ben Guerir as a share of the town's 2024 census count

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ABOUT THIS REPORT

Heirstone Consulting advises governments, developers, investors, mining companies and development finance institutions across the Middle East and Africa on markets, feasibility, land use and development strategy. This report draws on public sources current in 2026: national censuses and statistics offices, the World Bank, EITI and NRGI reports, company disclosures, government news agencies and the press.

Where we combine published figures to produce a new one, such as a growth multiple or a delivery ratio, the calculation is shown and labelled Heirstone analysis. The Democratic Republic of Congo has held no national census in more than 40 years, so Congolese towns appear in this report as references outside the panel.

WHAT WE FOUND

A mine can found a town. Open land and a second function decide whether it becomes a city.

We followed 25 African mining towns through two censuses each and compared their growth with their country's urban population. Mine size and mine life do not sort the towns. What separates the cities from the stranded towns is whether people can live and build outside the operator's fence, whether the town has a function that does not depend on the mine, and whether it was planned for the workforce that stays after construction.

1 **Most mining towns grow more slowly than their country's cities.**

Only 8 of the 25 towns in the Heirstone Mining Town Panel outgrew their national urban population over the same census years. Six are open towns, among them Solwezi in Zambia, host to First Quantum's Kansanshi mine since 2005, which grew 9.9% a year against 4.3% for urban Zambia. The other two are company towns that changed model: Oranjemund, opened to the public in 2017, and Ben Guerir, where OCP built a university.

2 **Fences hold towns still, even when the mine thrives.**

In Botswana, fenced Orapa lost 9% of its residents between 2011 and 2022 while Letlhakane, the open village next to it, grew 59%. Jwaneng, with about 20 years of mining ahead, grew 0.4% a year. All six towns that lost residents were company towns on our classification.

3 **The construction peak is the real urban shock.**

Simandou employed more than 60,000 workers at its construction peak and needs fewer than 15,000 to operate, according to Reuters, which also reported that 8,000 of 10,000 workers in the rail town of Dantilia lost their jobs within three months. Housing sized for the peak would lose most of its users once operations begin.

4 **Revenue shared on paper arrives in part.**

In the two cases where the amount due and the amount paid are both published, 37 to 38% of the amount due arrived: Katanga province and Ghana's Minerals Development Fund, both in 2014, with Ghana's figure on a cash basis that may mix years. Even paid in full, Ghana's local share is about 0.5% of gross mineral sales, and Zambia earmarks nothing for host areas.

5 **Towns outlive their mines through public functions.**

Kimberley became a provincial capital 80 years after the Big Hole closed and gained a university a century after. Lubumbashi and Kitwe are university cities. The newest conversions are planned: the closed Buzwagi gold mine at Kahama is becoming a special economic zone.

WHAT IT MEANS

Governments should zone open, titled land for the town before the mine is built, and publish each year how much of the host share arrived.

Mining companies should house the operating workforce permanently, treat the construction peak as temporary, and hand land and houses to residents well before closure.

DFIs and lenders should ask for a town land plan and a construction-to-operations workforce ratio alongside the environmental and social case.

Developers should favour open towns with a second function and underwrite to the operating workforce.

25

towns in nine countries in the Heirstone Mining Town Panel, with references in the DRC, Guinea, Mali and Mauritania

40+

years since the DRC last held a national census, so its mining cities sit outside the panel

CHAPTER 1

01

Most mines do not make cities

A mine brings jobs and wages, and often a town. Measured against the growth of their own country's cities, most African mining towns fall behind within a generation, including several beside the continent's most valuable mines.

Africa's mining towns are usually described one at a time: the boomtown, the company town, the ghost town after closure. To compare them we built the Heirstone Mining Town Panel: 25 towns in nine countries, each with two census counts from the same statistical series, set against the growth of the country's urban population over the same years. A town that grows faster than its country's cities is drawing people in. One that grows more slowly is losing ground, whatever its mine produces.

The result runs against the familiar picture of the mining boomtown. Only eight of the 25 towns outgrew their national urban population. Seventeen grew more slowly, and six of those lost residents outright: Orapa and Selebi-Phikwe in Botswana, Khouribga and Youssoufia in Morocco, and Redeyef and Moularès in Tunisia.

The towns that fell behind include some of the continent's most valuable mines. Debswana's Cut-8 and Cut-9 extension at Jwaneng supports mining for about another 20 years, with the pit expected to reach 816 metres by 2034. The town grew 0.4% a year between 2011 and 2022, a fifth of Botswana's urban rate. Khouribga, capital of Morocco's oldest phosphate field, counted 196,196 residents in 2014 and 195,931 in 2024.

Mine size and mine life do not sort the towns in our panel. The next chapters set out what does: whether the land is open, how the town is sized for construction and for operations, what share of mining revenue reaches it, and whether it has a function that would keep people there without the mine.

8 of 25

towns that grew faster than their national urban population

6

towns that lost residents between censuses, every one of them a company town

0.4%

annual growth of Jwaneng, 2011 to 2022, against 1.9% for urban Botswana

> *Only eight of 25 mining towns grew faster than their country's cities. All six that lost residents were company towns.*

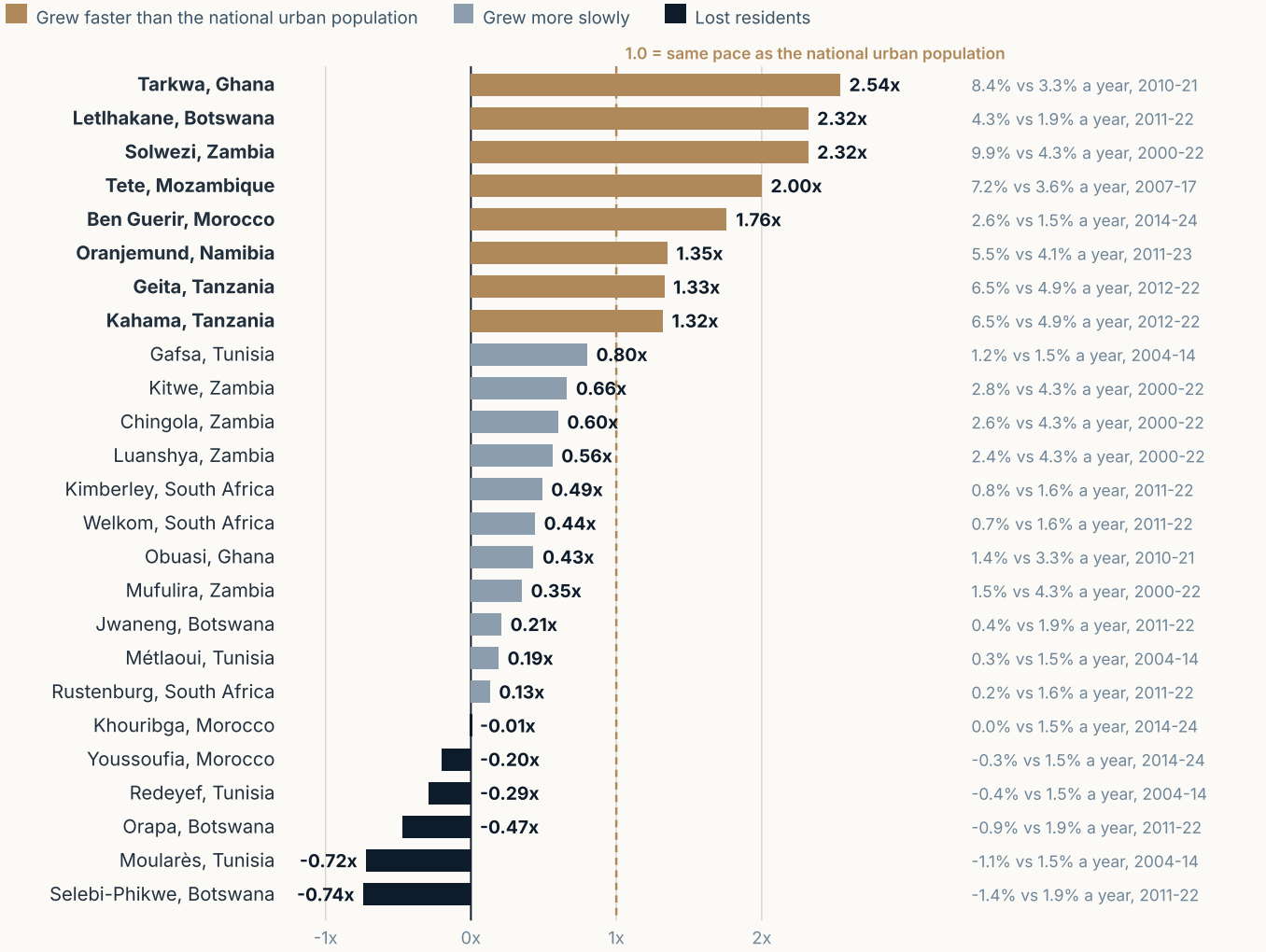
THE PANEL

Open towns lead the panel, and old company towns make up its tail

EXHIBIT 1

Only 8 of 25 African mining towns grew faster than their country's urban population

Town population growth as a multiple of national urban population growth over the same intercensal period (1.0 = same pace; below zero = the town lost residents). Right: town and national urban growth, % a year.



Source: Heirstone Mining Town Panel. Census counts compiled by City Population (Zambia, Morocco, Tunisia, Ghana); Statistics Botswana (2011, 2022 final; Letlhakane 2011 from the 2022 preliminary report); National Bureau of Statistics Tanzania; Stats SA, Census 2022 municipal fact sheet (2011 counts on 2021 boundaries); Oranjemund Town Council (2023 census) and Encyclopaedia Britannica (2011 census); INE Mozambique preliminary counts as reported by Encyclopaedia Britannica (2007) and Mozambique News Reports and Clippings (2018); World Bank WDI urban population (SP.URB.TOTL). **Heirstone analysis:** town compound annual growth divided by national urban compound annual growth. South African figures are local municipalities. Solwezi's 2022 count is compiled from ward data; Tarkwa's municipal boundaries may differ between censuses.

The fastest growers, Tarkwa, Letlhakane, Solwezi, Tete, Geita and Kahama, are open towns. Ben Guerir and Oranjemund are the two company towns among them, and both changed their model during the period measured. At the bottom sit the company towns of Botswana's diamond and copper-nickel fields and North Africa's phosphate basins.

Between them sits the largest group: the Zambian Copperbelt cities, South Africa's goldfields and platinum belt, and Obuasi. These are established mining cities that still grow, but more slowly than their countries. Kitwe added about 302,000 residents between 2000 and 2022 and still lost ground to urban Zambia.

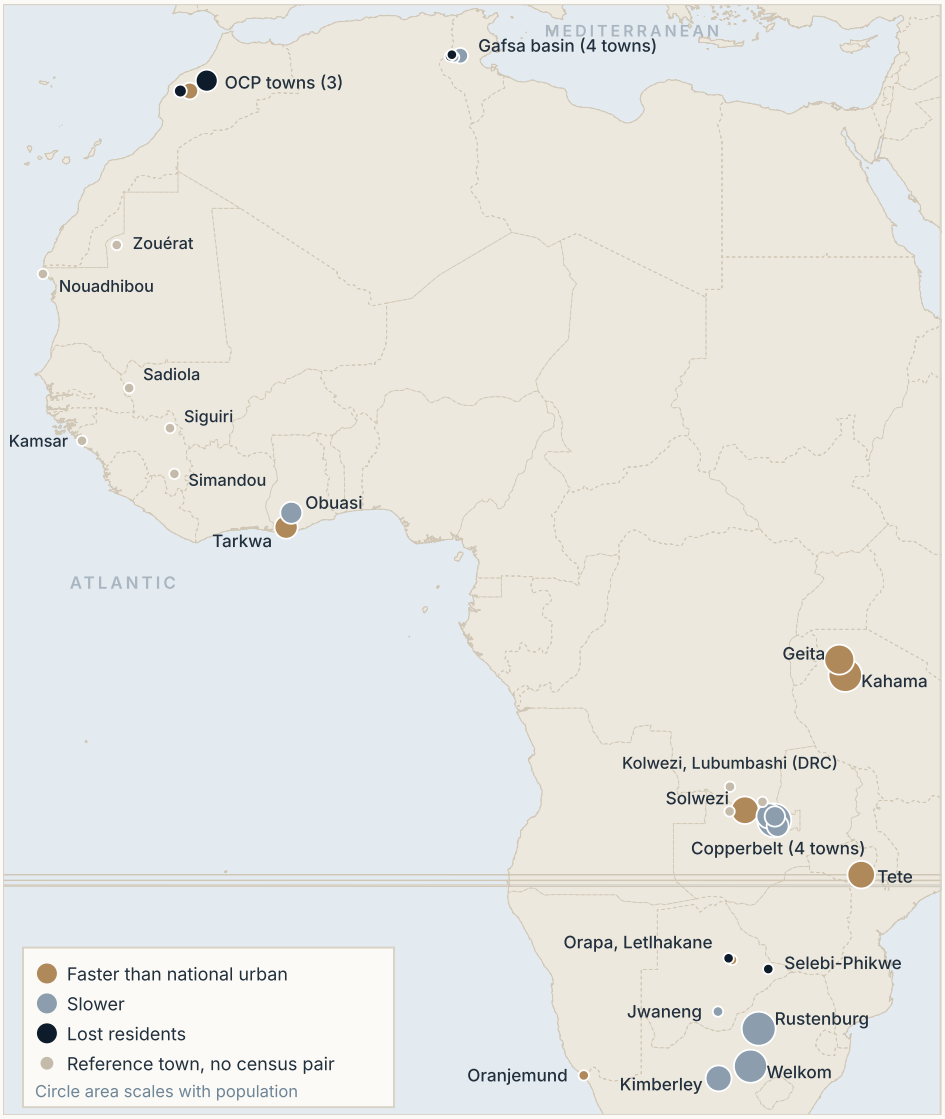
WHERE THE PANEL SITS

Nine countries and six commodities, from the Gafsa basin to the Namib coast

EXHIBIT 2

Fast growers cluster in the new copper and gold districts; the shrinking towns are diamond, copper-nickel and phosphate company towns

Heirstone Mining Town Panel by growth relative to the national urban population, latest intercensal period, with reference towns that lack a comparable census pair



Source: Heirstone Mining Town Panel (Exhibit 3); Natural Earth; town positions from the GeoNames gazetteer. Positions of Simandou (the range) and Kalumbila (at its airport) are approximate.

How the panel works

Each town has two census counts from the same statistical series, usually 10 to 12 years apart. We compare each town's compound annual growth with that of its country's urban population over the same years, from the World Bank's World Development Indicators. A multiple of 1.0 means the town kept pace with its country's cities.

What is left out

Kolwezi, Likasi and Lubumbashi, because the DRC has held no census in more than 40 years. Kalumbila and the Simandou corridor, because no earlier comparable count exists. Zouérat and Nouadhibou, because we could not obtain comparable town counts for both census rounds. Kathu, because Stats SA's 2022 count for its municipality is 29% below the 2011 count on the same boundaries, a fall the sources we reviewed do not explain.

Caveats

South African figures are local municipalities, which include non-mining areas. Boundaries can change between censuses. City Population compiles Zambia's 2022 figures from ward data that may be revised; Solwezi's 2022 count should be read with that in mind, although the town was already growing 7.0% a year between 1990 and 2010.

25
towns with two census counts from the same series

9
countries across North, West, East and Southern Africa

6
commodities: copper, gold, diamonds, phosphate, coal and platinum

9
reference towns without a census pair, in the DRC, Guinea, Mali, Mauritania and Zambia

THE BENCHMARK

The Heirstone Mining Town Panel

Towns grouped by Heirstone town type. **Company towns:** built by or for one operator to house its workforce and dominated by it as employer and landlord, even if since proclaimed a municipality. **Open boomtowns:** open land beside an active mine. **Legacy mining cities:** long-established towns where mining still sets the pace. **Diversified cities:** an administrative, port or university function carries the town. **Multiple:** town growth divided by national urban growth over the same years.

EXHIBIT 3

Every town that outgrew its country's cities was open or had changed model; every town that shrank was a company town

Census population, compound annual growth and growth multiple against the national urban population, by town type

TOWN	COUNTRY	COMMODITY	MINE OR TOWN	CENSUS YEARS	FIRST COUNT	LATEST COUNT	TOWN % A YEAR	URBAN % A YEAR	MULTIPLE
OPEN BOOMTOWNS									**MEDIAN 2.16X; 6 OF 6 FASTER**
Tarkwa	Ghana	Gold	University of Mines and Technology	2010-21	90,477	218,664	8.4	3.3	**2.54**
Letlhakane	Botswana	Diamonds	Open village beside Orapa	2011-22	22,911	36,404	4.3	1.9	**2.32**
Solwezi	Zambia	Copper, gold	Kansanshi, from 2005	2000-22	38,121	301,370	9.9	4.3	**2.32**
Tete	Mozambique	Coal	Moatize coal mines nearby	2007-17	152,909	305,722	7.2	3.6	**2.00**
Geita	Tanzania	Gold	Geita gold mine	2012-22	192,707	361,671	6.5	4.9	**1.33**
Kahama	Tanzania	Gold	Bulyanhulu; Buzwagi 2009-21	2012-22	242,208	453,654	6.5	4.9	**1.32**
COMPANY TOWNS THAT CHANGED MODEL DURING THE PERIOD									**MEDIAN 1.56X; 2 OF 2 FASTER**
Ben Guerir	Morocco	Phosphate	OCP; UM6P from 2017	2014-24	88,626	114,872	2.6	1.5	**1.76**
Oranjemund	Namibia	Diamonds	Namdeb; opened to public 2017	2011-23	3,908	7,441	5.5	4.1	**1.35**
DIVERSIFIED CITIES									**MEDIAN 0.65X; NONE SHRANK**
Gafsa	Tunisia	Phosphate	CPG; governorate capital	2004-14	84,676	95,242	1.2	1.5	**0.80**
Kimberley	South Africa	Diamonds	Big Hole closed 1914	2011-22	248,041	270,078	0.8	1.6	**0.49**
LEGACY MINING CITIES									**MEDIAN 0.44X; NONE FASTER**
Kitwe	Zambia	Copper	Copperbelt	2000-22	363,734	665,961	2.8	4.3	**0.66**
Chingola	Zambia	Copper	Copperbelt	2000-22	147,448	256,560	2.5	4.3	**0.60**
Luanshya	Zambia	Copper	Copperbelt	2000-22	115,579	193,293	2.4	4.3	**0.56**
Welkom	South Africa	Gold	Free State goldfield	2011-22	407,020	439,034	0.7	1.6	**0.44**
Obuasi	Ghana	Gold	AngloGold Ashanti	2010-21	168,641	196,698	1.4	3.3	**0.43**
Mufulira	Zambia	Copper	Copperbelt	2000-22	122,336	169,136	1.5	4.3	**0.35**
Rustenburg	South Africa	Platinum	Platinum belt	2011-22	549,575	562,315	0.2	1.6	**0.13**
COMPANY TOWNS									**MEDIAN -0.24X; 6 OF 8 LOST RESIDENTS**
Jwaneng	Botswana	Diamonds	Debswana, from 1982	2011-22	18,008	18,784	0.4	1.9	**0.21**
Métlaoui	Tunisia	Phosphate	CPG	2004-14	37,099	38,129	0.3	1.5	**0.19**
Khouribga	Morocco	Phosphate	OCP	2014-24	196,196	195,931	0.0	1.5	**-0.01**
Youssoufia	Morocco	Phosphate	OCP	2014-24	67,628	65,669	-0.3	1.5	**-0.20**
Redeyef	Tunisia	Phosphate	CPG	2004-14	26,143	25,046	-0.4	1.5	**-0.29**
Orapa	Botswana	Diamonds	Debswana closed town	2011-22	9,531	8,648	-0.9	1.9	**-0.47**
Moularès	Tunisia	Phosphate	CPG	2004-14	23,841	21,431	-1.1	1.5	**-0.72**
Selebi-Phikwe	Botswana	Copper-nickel	BCL, closed 2016	2011-22	49,411	42,488	-1.4	1.9	**-0.74**

Source: As Exhibit 1. Town types are a **Heirstone classification** based on published descriptions of each town's housing and administrative status. Kimberley, Welkom and Rustenburg are the Sol Plaatje, Matjhabeng and Rustenburg local municipalities; Obuasi combines Obuasi Municipal and Obuasi East. Khouribga and Youssoufia are also provincial seats; we class them by their dominant employer.



CHAPTER 2

02 Behind the fence

Many of Africa's older mining towns were built as company towns. In the panel they include the phosphate towns of OCP in Morocco and of the Compagnie des Phosphates de Gafsa in Tunisia. Debswana describes Orapa as a closed town, and Oranjemund, established in 1936 in the diamond fields, opened to the public only in 2017.

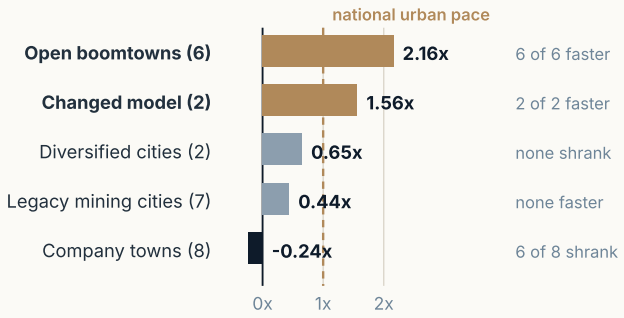
The model suited a remote mine. The operator decided who lived in town, provided houses, schools and clinics, and planned for its own workforce. It also tied the size of the town to the size of the payroll. When the payroll stopped growing, so did the town, and when mechanisation cut jobs the town shrank.

Exhibit 4 groups the panel by town type. The eight company towns have a median multiple below zero, and six of them lost residents. Open boomtowns grew at a median 2.2 times their country's urban rate. The two company towns that changed model during the period, one by opening its gates and one by adding a university, both outgrew their countries' cities.

EXHIBIT 4

Company towns stood still while open boomtowns grew about 2.2 times as fast as their countries' cities

Median growth multiple against the national urban population, by Heirstone town type; number of towns in brackets



Source: Heirstone Mining Town Panel (Exhibit 3). **Heirstone analysis.**

6 of 8

company towns in the panel that lost residents between censuses

0 of 8

company towns that grew faster than their country's cities without changing model

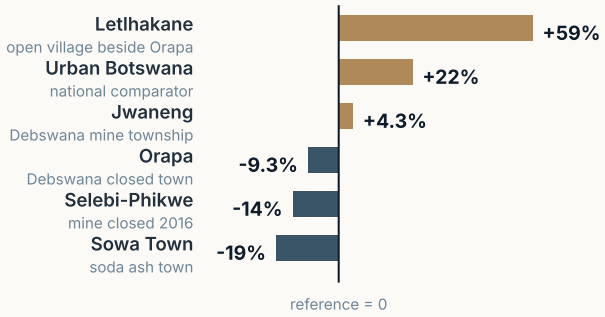
DIAMOND TOWNS

In Botswana, the open village next door took the growth that the company town did not

EXHIBIT 5

Orapa lost 9% of its residents while open Letlhakane grew 59%

Change in census population, 2011 to 2022, %

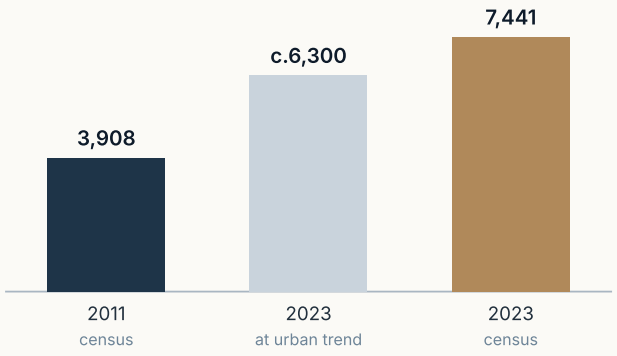


Source: Statistics Botswana, Population and Housing Census 2011 and 2022 (final counts; Letlhakane 2011 count from the 2022 preliminary results report); World Bank WDI. **Heirstone analysis.**

EXHIBIT 6

Oranjemund almost doubled after it opened, ending about 1,100 residents above Namibia's urban trend

Census population, 2011 and 2023, and the 2023 population implied by urban Namibia's growth



Source: Encyclopaedia Britannica (2011 census); Oranjemund Town Council (2023 census); World Bank WDI. **Heirstone analysis:** trend applies urban Namibia's 2011-2023 growth of 61.8% to the 2011 count. Oranjemund was proclaimed a town in 2011 and opened to the public in 2017 (Oranjemund Town Council).

Orapa shows the pattern most plainly. Debswana describes it as a closed town of over 12,000 people serving the Orapa, Letlhakane and Damtshaa mines, and says the Orapa mine averages about 10 million carats a year. The census, which defines the town differently, counted 9,531 residents in 2011 and 8,648 in 2022. Over the same years Letlhakane, the open village next to Orapa and the Letlhakane mine, grew from 22,911 to 36,404, more than twice as fast as urban Botswana. On these figures, the population growth around the Orapa field went to the open village.

Jwaneng shows that mine life does not change the pattern. Debswana built the township together with the mine, which became fully operational in 1982 and has about 20 years of extension ahead, yet the town grew 0.4% a year between 2011 and 2022.

Namibia ran the experiment the other way. Oranjemund, established in 1936 after alluvial diamonds were found near the Orange River, was proclaimed a town in 2011 and opened to the public in 2017, according to the town council. Its population rose from 3,908 at the 2011 census to 7,441 at the 2023 census. That is almost double, and about 1,100 residents more than urban Namibia's growth over the same years would have produced.

-9%

Orapa, 2011 to 2022, beside a mine producing about 10 million carats a year

+59%

Letlhakane, the open village next door, over the same years

1.9x

Oranjemund's population, 2011 to 2023, after it was proclaimed and opened

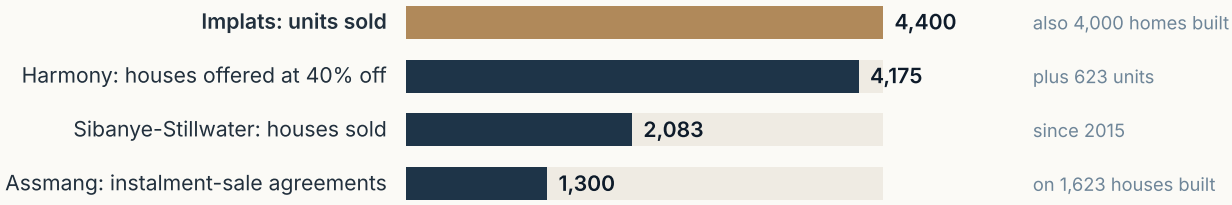
HOUSING

South Africa's miners are selling company houses to the people who live in them

EXHIBIT 7

Four producers have sold, contracted or offered close to 12,000 homes to employees

Mine housing transferred or offered to employees, by company, as reported in October 2024



Source: Minerals Council South Africa, first accommodation newsletter, via Mining Weekly (21 October 2024). Harmony figure is housing stock on offer, not completed sales. **Heirstone analysis** for the total.

South Africa's mining towns grew around hostels and company houses. The Mineral and Petroleum Resources Development Act of 2002 and the Mining Charter now require each mine to agree Social and Labour Plans, and housing is a large part of them. In October 2024 the Minerals Council published what the larger producers had done. Impala Platinum reported spending more than R5bn on housing and living conditions (about US$270m at the 2024 average of R18.33 to the dollar), building 4,000 homes, selling 4,400 units and converting hostels into 5,243 single and 308 family units.

Harmony is selling its stock of 4,175 houses and 623 units to employees at a 40% discount. Sibanye-Stillwater has sold 2,083 houses since 2015, about 260 a year. Assmang has built 1,623 houses on 2,150 serviced stands, with 1,300 instalment-sale agreements. Around 2013 Anglo American reported 1,662 houses built for Sishen employees at Kathu, with 1,720 more planned.

Selling the house to the worker turns a liability on the mine's books into a household asset, and gives the town a property market that does not depend on one payroll. Once a house is sold, its later use no longer depends on the mine's staffing.

Elsewhere the boundary between mine and town is still being drawn. In Kolwezi, AFP reported in 2022 that the Gécamines quarter, a district of 38,000 residents with reliable electricity and running water, could disappear within three years as a neighbouring mine expanded, and that about 2,000 people had left in six months. For investors in the city, the case shows the value of a published boundary between mining titles and urban land.

R5bn+

spent by Implats on housing and living conditions, with 4,400 units sold

40%

discount at which Harmony is offering mine houses to employees

38,000

residents of Kolwezi's Gécamines quarter when AFP reported its likely relocation (2022)

CHAPTER 3

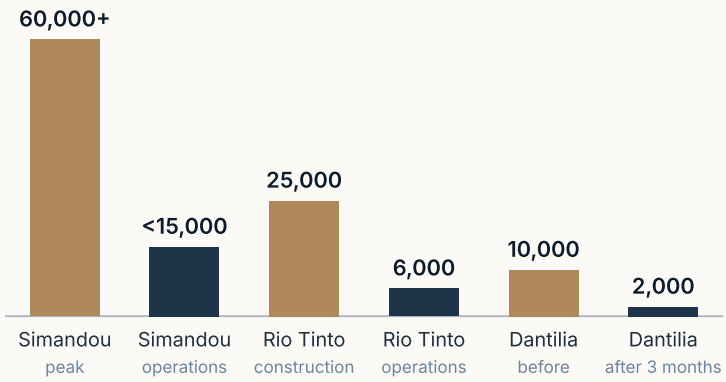
03 The construction cliff

A large mine needs several times more people to build than to run. The town feels the construction peak first, and the drop a few years later.

EXHIBIT 8

Simandou needs about a quarter of its construction workforce to operate

Workers at construction peak and needed for operations; Dantilia, workers before and three months into demobilisation, late 2025



Source: Reuters via Mining.com (18 December 2025). Simandou operations figure is the stated maximum for mines, railway and port. Dantilia figures as reported by Reuters. **Heirstone analysis** for ratios.

4 to 1

ratio of Simandou's construction peak to its operating workforce

Simandou is an iron-ore mine, railway and port built as one system. SimFer's part of the project alone needs an estimated US$11.6bn of capital; the corridor includes about 600 km of new railway and a port at Morebaya, and can carry up to 120 million tonnes a year. The first cargo was loaded in November 2025.

Construction employed more than 60,000 people at its peak in 2024 and 2025, according to Reuters. Running the mines, railway and port needs fewer than 15,000. Rio Tinto put its own construction workforce at about 25,000, 82% Guinean, against about 6,000 for operations. As work moved from building to running, Reuters reported, 8,000 of the 10,000 workers in the rail town of Dantilia lost their jobs within three months, and about 1,500 in Kamara.

The project expects 10,000 direct and more than 100,000 induced jobs in operation, according to SimFer figures reported by Mining Weekly. That is a lasting base for corridor towns, on a different timetable: housing and services built for the peak can lose three quarters of their users within months, while induced jobs build up over years.

In the sources we reviewed we found no published town or housing plan for the corridor. Rio Tinto Simfer's 2023 community package in Beyla, GNF 44bn (about US$5.2m at GNF 8,511, the World Bank's 2023 factor), funds 15 health, education and agrifood projects and no housing.

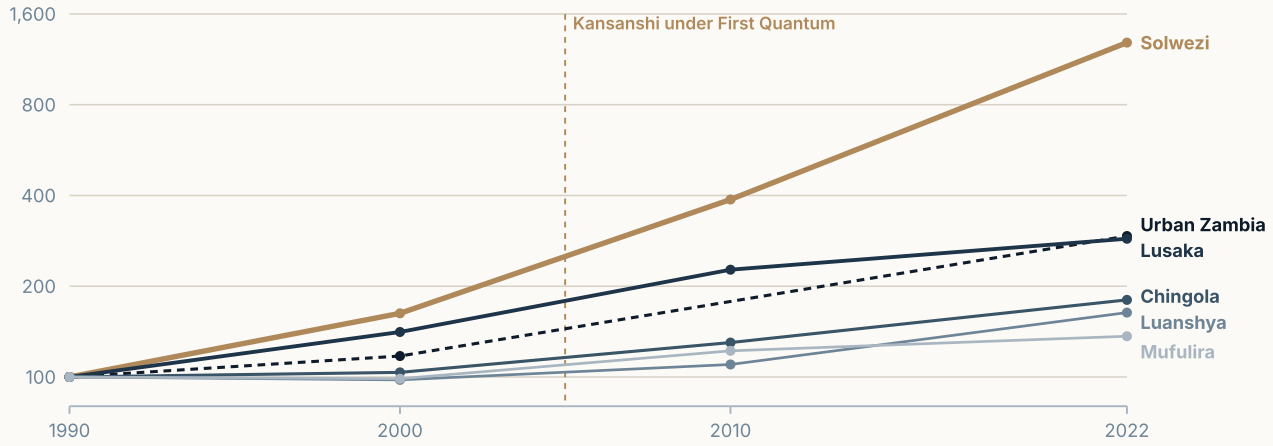
THE NEW COPPERBELT

Solwezi's population grew faster than its streets and services

EXHIBIT 9

Solwezi grew almost 13-fold after 1990; the old Copperbelt towns trailed urban Zambia

Population index, 1990 = 100, census years 1990, 2000, 2010 and 2022 (logarithmic scale)



Source: City Population compiling Zambia Statistics Agency censuses (2022 figures compiled from ward data); World Bank WDI urban population (no 2010 point plotted). **Heirstone analysis.**

Solwezi had 23,435 residents in 1990. First Quantum has run the Kansanshi mine, 10 km north of the town, as its flagship since 2005, and the town also hosts the Lumwana mine. Solwezi counted 90,856 people in 2010 and, on City Population's compilation of ward data, 301,370 in 2022. Kansanshi employs more than 13,000 people, and its S3 expansion produced first ore in July 2025. In 2015 most of Solwezi's roads still had no names, according to the Times of Zambia.

The privatisation decade shows the reverse. Between 1990 and 2000, as ZCCM was broken up and sold, Luanshya and Mufulira lost residents and Chingola barely grew, while Lusaka grew 41%. The old Copperbelt recovered after 2000 but did not catch up.

Kalumbila, 150 km west of Solwezi, is the planned alternative. First Quantum built the Sentinel mine there from 2012 for US$2.1bn. Its January 2014 presentation planned up to 10,000 homes in a new town of about 50,000 residents, most of them to be built by third parties. By December 2017 about US$80m had gone on housing, according to Zambia's Daily Nation, about 3.8% of the mine's build cost. The district counted 177,067 people in the preliminary results of the 2022 census.

EXHIBIT 10

Kalumbila planned up to 10,000 homes, most of them for third parties to build

Homes at Kalumbila new town, Sentinel copper mine



Source: First Quantum, Trident presentation (January 2014), for the plan and phases; Lusaka Times (12 August 2013); Daily Nation, Zambia (6 December 2017), for the Phase 2 count reported by the press. District population: Zambia Statistics Agency, 2022 Census of Population and Housing, Preliminary Report; district created in 2015 (Lusaka Times, 2015).

3.8%

of Sentinel's US$2.1bn build cost spent on Kalumbila housing by December 2017 (about US$80m)

177,067

residents of Kalumbila district in 2022, seven years after it was created

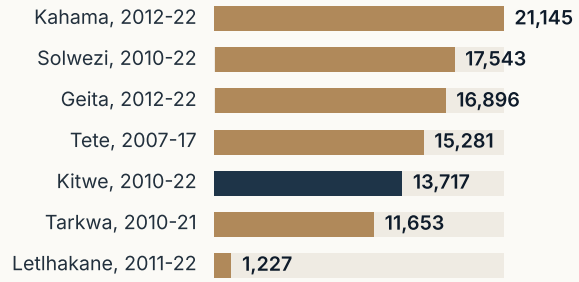
ABSORPTION

At the height of a boom, the larger open towns took in about 12,000 to 21,000 new residents a year

EXHIBIT 11

Five open boomtowns each added about 12,000 to 21,000 residents a year

Average annual increase in census population, latest intercensal period; Kitwe shown as a legacy city reference

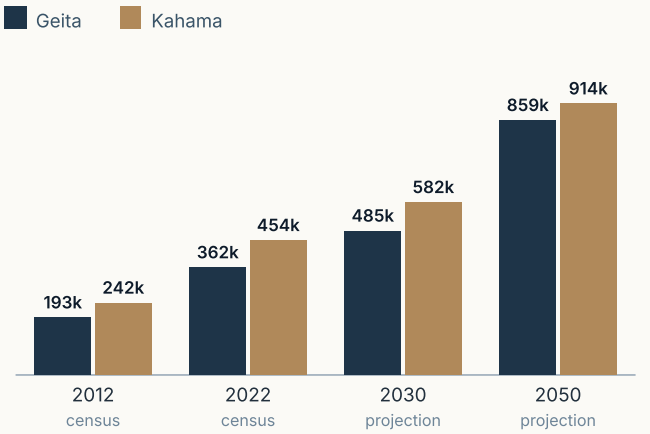


Source: NBS Tanzania; City Population (Zambia, Ghana); INE Mozambique preliminary counts (Encyclopaedia Britannica; Mozambique News Reports and Clippings, 2018); Statistics Botswana. Tarkwa municipal boundaries may differ between censuses; Solwezi's 2022 count is compiled from ward data. **Heirstone analysis.**

EXHIBIT 12

Tanzania's statistics office expects Geita and Kahama to reach about 0.9 million people each by 2050

Population of Geita Town Council and Kahama Municipal Council, thousands



Source: National Bureau of Statistics Tanzania, council population projections (2022 census base).

Tanzania's two gold towns show the scale. Geita Town Council grew from 192,707 to 361,671 between 2012 and 2022, 6.5% a year compounded; the National Bureau of Statistics gives 7.4%. Kahama Municipal Council, home to Bulyanhulu and until 2021 Buzwagi, grew from 242,208 to 453,654. Together they added about 380,000 people in a decade, and the bureau projects 858,573 for Geita and 914,350 for Kahama by 2050.

Both towns gained functions as they grew. Geita is now the seat of its own region. At Kahama, Barrick funded a new airport terminal for more than 200 passengers.

Tete, on the Zambezi near the coal mines of Moatize, doubled from 152,909 to 305,722 between the preliminary counts of the 2007 and 2017 censuses.

A town adding 15,000 to 20,000 people a year needs serviced land, water connections and school places on that scale every year. Company housing programmes run to hundreds of homes a year, as at Sibanye-Stillwater, or a few thousand in total, as at Implats. The rest of the market is built by households and small landlords, on whatever land the town can release.

21,145

average annual increase in Kahama's population, 2012 to 2022

380,000

residents added by Geita and Kahama together in one decade

CHAPTER 4

04

Money due, money paid

37%

of the provincial and local royalty share due under the law reached Katanga in 2014. Ghana's Minerals Development Fund received about 38% of its share in the same year.

Several African mining codes send part of mining revenue to the province, district or community that hosts the mine. Where the amount due and the amount paid are both published, a little over a third arrived.

The DRC's 2018 Mining Code allocates 25% of the mining royalty to the host province, 15% to the local territorial entity hosting the project and 10% to a fund for future generations, and requires companies to put 0.3% of turnover into community development each year. Under the 2002 code the provincial and local share was 40%. In 2014 Katanga declared receiving US$12.2m against US$33.4m owed, according to a 2017 briefing by the Natural Resource Governance Institute: 37% of its entitlement.

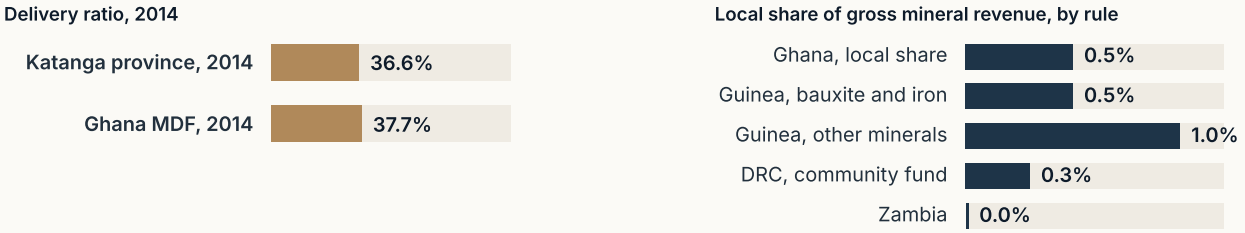
Ghana allocates 20% of its 5% mineral royalty to the Minerals Development Fund, and 10% of royalties go to district assemblies, stools and traditional councils through the Office of the Administrator of Stool Lands. In 2014 Ghana collected GHS 469m in royalties, about US$162m at that year's average of GHS 2.90 to the dollar. GHEITI reported MDF disbursements of GHS 35.4m, 7.5% of royalties against the 20% due, or about 38% of the entitlement, on a cash basis that may include earlier years' amounts.

Both cases are more than a decade old. The DRC has since moved to direct payment by companies to local entities, from July 2018. EITI found that between 2016 and 2018 revenue was not passed to the local level as the rules required. According to AFREWATCH, debit notes showed US$35.2m of royalty due to Congolese local entities for January to May 2021, but the payments were not published. In July 2026 GHEITI urged Ghana's Parliament to enforce delayed disbursements.

EXHIBIT 13

Where both sides of the ledger are published, 37 to 38% of the amount due arrived, and the share itself is under 1% of sales

Left: Heirstone delivery ratio, subnational or fund revenue received as a share of the statutory entitlement, 2014. Right: statutory local share as a percentage of gross mineral revenue.



Source: NRGI, Iwerks and Toroskainen (April 2017); GHEITI, 2014 Mining Sector Report; EITI country pages for Ghana, Guinea, DRC and Zambia (accessed 2026). **Heirstone analysis.** Ghana ratio is on a cash basis and may include timing differences (GHEITI). Right panel: Ghana, 10% of a 5% royalty; Guinea, company contribution on turnover; DRC, 0.3% community fund only, as royalty rates by substance were not verified; Zambia earmarks none.

Even paid in full, statutory shares are small beside a growing town's needs. Ghana's local share works out at about 0.5% of gross mineral sales, so a mine selling US$1bn of gold a year would yield about US$5m for local recipients.

According to EITI, mining royalties and taxes still make up about 40% of the budgets of the local governments that receive them.

0.5%

of gross mineral sales: Ghana's statutory local share when paid in full

THE RULES

Zambia earmarks nothing for host areas and Guinea shares its local fund across the whole country

EXHIBIT 14

Six countries route mining revenue to host areas in different ways, and few payments are published

Statutory and contractual channels from mining revenue to host areas, with latest published amounts

COUNTRY	INSTRUMENT	SHARE AND BASE	HOST AREA ONLY?	LATEST PUBLISHED AMOUNT	WHAT THE RECORD SHOWS
DRC	Royalty split, Mining Code 2018	25% province, 15% local entity, 10% future-generations fund	Yes	US$35.2m due to local entities, Jan-May 2021; payments not published	Katanga received 37% of its due in 2014; EITI found non-disbursement 2016-18
DRC	Community development fund	0.3% of turnover a year	Yes	Not published	Handbook only in December 2021; 11 of 52 producers disclosed social spending
DRC	Provincial levies	Various	By province	US$528.7m in 2020-21, 62.8% in Lualaba	Reported by ITIE-RDC
Ghana	Royalty share, MDF Act 2016	20% of royalties to MDF; 10% to districts, stools and councils (55/25/20)	Yes	GHS 35.4m disbursed, 2014 (US$12.2m)	7.5% of royalties paid against 20% due; royalties and taxes about 40% of local budgets
Guinea	National Local Development Fund	15% of mining tax revenue	No: all local authorities	GNF 430bn in revised 2023 budget (US$50.5m), cut from GNF 570bn (US$67.0m)	Cut by a quarter in September 2023
Guinea	Local development contribution	0.5% of turnover (bauxite, iron); 1% other minerals	Yes	Not found	
Guinea	Surface royalty	90% to communes, 10% to prefectures	Yes	Not found	
Zambia	None earmarked	All to the consolidated fund	No	Mining taxes US$1.9bn, 2022	44% of government revenue
South Africa	Social and Labour Plans (company duty)	Company-defined five-year plans	Yes	Social investment R4.9bn, 2023 (US$266m)	3.6% of the R135bn paid to the fiscus
Morocco	No statutory local share identified	Corporate investment by OCP	Host town	SADV capital MAD 2.6bn (US$261m)	Green City for 100,000 residents

Source: EITI country pages (DRC, Ghana, Guinea, Zambia, accessed 2026); NRGI (April 2017); AFREWATCH (April 2022); Infos.cd citing ITIE-RDC (17 April 2023); GHEITI 2014 Mining Sector Report; Guinéenews (15 September 2023); Minerals Council South Africa via Modern Mining (20 June 2024); SADV (2020). Conversions at World Bank annual averages: GHS 2.90 (2014), GNF 8,511 (2023, DEC alternative conversion factor), R18.45 (2023). Botswana, Namibia, Tanzania and Tunisia rules were not verified.

Zambia sits at one end. Mining taxes reached US$1.9bn in 2022, 44% of government revenue and 72% of export earnings, and all of it goes to the consolidated fund. No share is earmarked for Solwezi, Kalumbila or the Copperbelt districts.

Guinea's 15% National Local Development Fund is shared among all the country's local authorities, not only mining communes. The 2023 budget gave it GNF 570bn; the September revision cut it to GNF 430bn, from about US$67m to US$51m. The mining investment fund was cut in the same proportion.

Host communes rely more directly on the 0.5% turnover contribution owed by bauxite and iron-ore producers, for which we found no published payments.

In South Africa the duty sits with companies. The Minerals Council estimated industry social investment at R4.9bn in 2023, 3.6% of the R135bn the sector paid in taxes and royalties. Social and Labour Plans are the larger channel: Sibanye-Stillwater spent R13.5bn on 15 plans between 2013 and 2023, against R472m of discretionary social spending.

US$1.9bn

Zambian mining taxes in 2022, none earmarked for host areas

-25%

cut to Guinea's National Local Development Fund in the September 2023 revised budget

US$332m

collected from extractives by Lualaba's provincial revenue directorate in 2020-21, on ITIE-RDC figures



CHAPTER 5

05

When the mine stops

None of the closures we reviewed emptied its town. Each removed the payroll that had set the town's size, and the population then adjusted to what other employers and public functions could support.

Selebi-Phikwe is the most closely documented case. BCL, its copper-nickel miner, was placed in provisional liquidation in October 2016, and the contracts of its 5,500 workers and 700 at Tati Nickel were to be terminated. Botswana's government news agency put the effect at about 5,000 employees and 20,000 dependents. BCL had contributed 2.4% of the country's GDP.

The census counted 49,411 residents in 2011 and 42,488 in 2022, a fall of 14%. The town lost about 1.3 residents for every BCL job. Against Botswana's urban growth over the same years, the shortfall is larger, at about three residents per job (Exhibit 15).

Welkom shows a slower version. It sits in Matjhabeng, a municipality where about ten gold mines have closed shafts, according to a municipal spokesperson quoted by News24 in 2021. On Stats SA census counts, Matjhabeng still grew from 407,020 to 439,034 between 2011 and 2022, at less than half the pace of urban South Africa. South African gold mining employed 93,841 people in 2022 and produced 84 tonnes.

> *Selebi-Phikwe lost about 1.3 residents for every BCL job. Measured against Botswana's urban trend, it lost about three.*

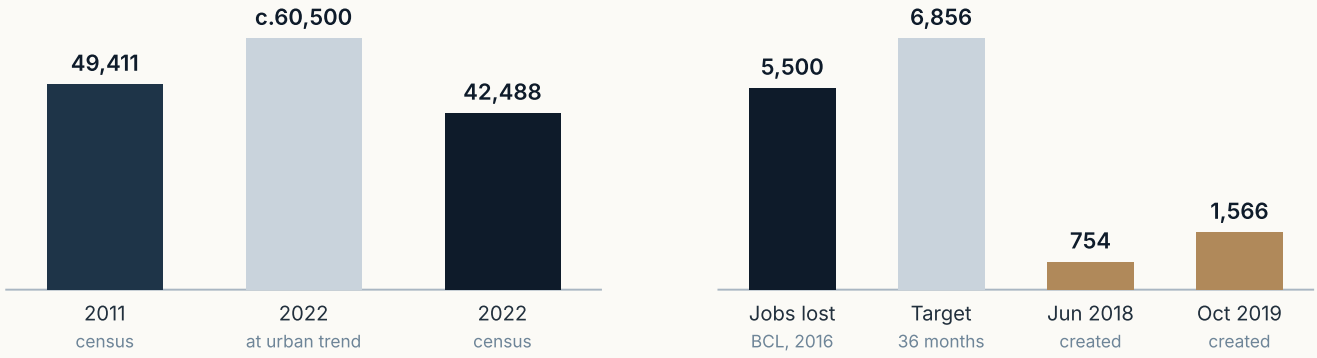
SELEBI-PHIKWE

Three years after BCL closed, a dedicated agency had replaced about a quarter of the jobs

EXHIBIT 15

Selebi-Phikwe is about 18,000 people short of its trend, and replacement jobs reached a quarter of those lost

Left: census population and the 2022 population implied by Botswana's urban growth, 2011 to 2022. Right: BCL jobs lost, SPEDU job target and jobs reported created



Source: Statistics Botswana (2011 and 2022 censuses); World Bank WDI; IndustriALL (October 2016); The Patriot on Sunday (c.2017); Botswana Daily News (27 June 2018; 1 October 2019). **Heirstone analysis:** trend applies urban Botswana's 2011-2022 growth of 22.4% to the 2011 count.

SPEDU, the regional development agency for the town, aimed to create 6,856 jobs within 36 months of the closure, according to The Patriot on Sunday, from projects worth P1.185bn (about US$114m at the 2017 average of BWP 10.35 to the dollar). By June 2018 Botswana's government news agency reported 754 jobs created; by October 2019, about 1,566, with seven citizen-owned companies. That is 26 to 31% of the 5,000 to 6,000 jobs lost, and about 23% of the target. Kimberley's capital status and university came 80 and 100 years after its mine closed.

The mine itself may return. NexMetals, formerly Premium Resources, is redeveloping the past-producing Selebi mines and in 2026 reported an inferred resource at Selebi Main of 32.47 million tonnes at 1.53% copper and 0.98% nickel, 72% more than before. A restart would find the town's housing and services still in place.

1.3
residents lost per BCL job, 2011 to 2022 census counts

26-31%
of BCL-related jobs replaced by October 2019, three years after closure

23%
of SPEDU's 36-month job target reached by October 2019

CONVERSION

At Kahama, a closed gold mine is being handed over as an economic zone

Barrick closed the Buzwagi gold mine at Kahama in July 2021. The site is being converted into the Buzwagi Special Economic Zone, which a pre-feasibility study expects to create at least 3,000 jobs a year, and it hosts the Barrick Academy training centre. More than 11 investors had expressed interest by mid-2025, according to Mining Weekly. In March 2026 a parliamentary committee recommended moving management of the zone from Kahama Municipal Council to TISEZA, the national special economic zones authority.

Buzwagi is useful as a template because of its sequence: the mine site and a training centre pass to a public zone whose job is to attract other employers, and the town has gained new airport capacity. Kahama kept growing through the closure, to 453,654 residents in 2022, with Bulyanhulu still operating nearby.

In the DRC, the Musompo Special Economic Zone near Kolwezi applies the idea well before closure. Works began in March 2025 on more than 900 hectares, at a development cost of over US$200m, with a target of nearly US$2bn of investment in battery precursors and batteries and about 25,000 direct jobs, according to Bankable Africa.

3,000

jobs a year expected from the Buzwagi zone in its pre-feasibility study

900+ ha

Musompo Special Economic Zone, Lualaba, where works began in March 2025

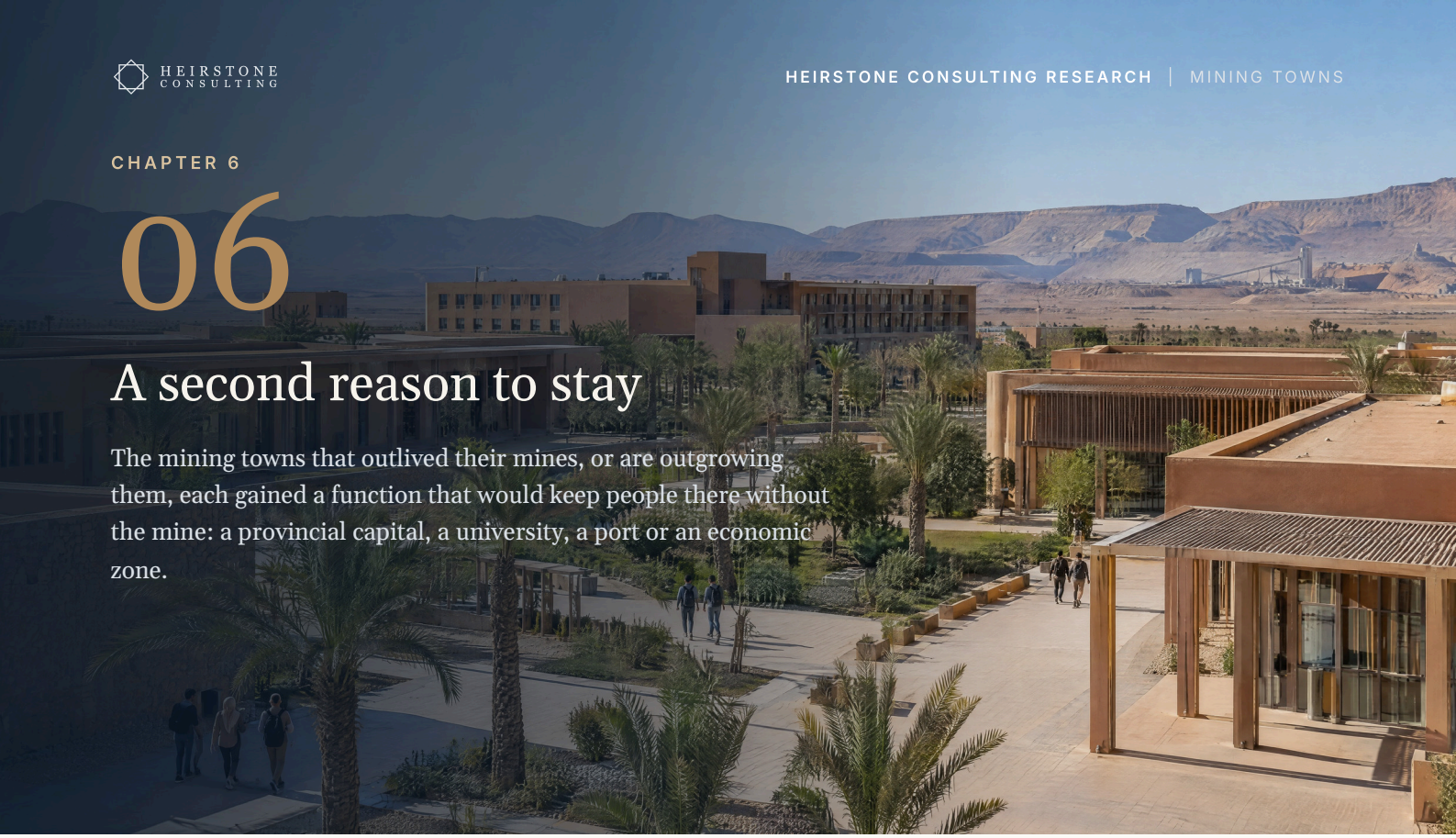
EXHIBIT 16

A century of closures and sales: none of the towns listed has emptied

Selected closure, ownership and conversion events in African mining towns

YEAR	TOWN	EVENT	JOBS AFFECTED	WHAT FOLLOWED
1914	Kimberley, South Africa	Mining at the Big Hole stops	n/p	Provincial capital 1994; Sol Plaatje University 2014
to 2000	Zambian Copperbelt	ZCCM's operating divisions privatised	n/p	Luanshya and Mufulira shrank in 1990-2000 while Lusaka grew 41%
2011-2022	Gafsa basin, Tunisia	Commercial phosphate output at most 3.5 Mt a year, against 8 Mt in 2010	n/p	Redeyef and Moularès shrank, 2004-14
2011, 2017	Oranjemund, Namibia	Company town proclaimed, then opened to the public	n/a	Population 3,908 (2011) to 7,441 (2023)
2016	Selebi-Phikwe, Botswana	BCL in provisional liquidation	5,500 BCL and 700 Tati jobs	Population -14%, 2011-22; SPEDU; Selebi redevelopment studies
2024	Zambian Copperbelt	51% of Mopani, operator of the former Nkana and Mufulira divisions, sold	n/p	Buyer International Resources Holding, based in Abu Dhabi, paid about US$1.1bn
2021	Kahama, Tanzania	Buzwagi gold mine closes	n/p	Buzwagi SEZ; Barrick Academy; zone authority management proposed 2026
to 2021	Welkom, South Africa	About ten gold mines close shafts	n/p	Municipality +8%, 2011-22; mine houses sold to employees

Source: Heirstone closure events register, compiled from IndustriALL (2016); Mining Weekly (25 July 2025); AllAfrica/Daily News (16 March 2026); Statistics Botswana; City Population; Stats SA; South African History Online; Sol Plaatje University; ZCCM-IH (2021); Forbes Middle East (2024); AGBI (2023); Oranjemund Town Council; Encyclopaedia Britannica; News24 (2021). n/p: not published.



CHAPTER 6

06

A second reason to stay

The mining towns that outlived their mines, or are outgrowing them, each gained a function that would keep people there without the mine: a provincial capital, a university, a port or an economic zone.

Ben Guerir is the panel's best example of a miner building a second function before the town needs one. OCP mines phosphate beside the town. Construction of the Mohammed VI Green City began in June 2008, according to a 2010 press report, as a 650 ha city for 90,000 residents. The plan has since grown to 1,000 ha for 100,000 inhabitants, with 25,000 homes, a 188 ha technology park, 200,000 m^2 of retail space and Mohammed VI Polytechnic University (UM6P) at its centre.

UM6P, at the core of the Green City, was inaugurated in 2017, according to the official news agency MAP. The developer, SADV, is a wholly owned OCP subsidiary created in 2011 with share capital of MAD 2.6bn, which SADV states as US$261m. Its first urban core was to exceed 350 ha by 2025, and UM6P targeted 6,000 students and researchers by that year and 20,000 at full development.

The town has grown with it. Ben Guerir was already outpacing urban Morocco in 2004-2014, after Green City works began and before UM6P opened. It grew from 62,872 in 2004 to 114,872 in 2024, 3.1% a year, while Khouribga grew 0.8% a year and Youssoufia barely grew over the same two decades. Urban Morocco grew 1.9% a year. Once complete, the Green City alone would house the equivalent of 87% of the town's 2024 population.

87%

planned Green City population as a share of Ben Guerir's 2024 census count

3.1%

annual growth of Ben Guerir, 2004 to 2024, against 0.8% for Khouribga

SECOND FUNCTIONS

Capitals, universities and economic zones carry mining towns past the mine

EXHIBIT 17

Nine mining towns and the functions they added

Non-mining public or institutional functions in selected mining towns, with the year added where published

TOWN	MINING FROM	SECOND FUNCTION	ADDED	SCALE OR OUTCOME
Kimberley	Diamonds; Big Hole closed 1914	Provincial capital; Sol Plaatje University	1994; 2014	Municipality 270,078 (2022)
Lubumbashi	1910, copper	University; international airport	1955	No census in more than 40 years
Kitwe	Copper	Copperbelt University	1987	Grew 0.66 times urban Zambia's pace
Tarkwa	Gold	University of Mines and Technology	n/p	Grew 2.5 times urban Ghana's pace
Ben Guerir	Phosphate, OCP	UM6P; Green City; 188 ha technology park	2017	6,000 students and researchers targeted for 2025
Solwezi	2005, Kansanshi	Capital of North-Western Province; airport	n/p	9.9% a year, 2000-22
Geita	Gold	Regional headquarters	n/p	6.5% a year, 2012-22
Kolwezi	Copper and cobalt	Musompo special economic zone	2025	Over 900 ha; about 25,000 direct jobs targeted
Kahama	Buzwagi 2009-21; Bulyanhulu	Buzwagi SEZ; Barrick Academy; airport terminal	2021-26	At least 3,000 jobs a year expected

Source: Heirstone Mining Town Panel; First Quantum; Mining Weekly (25 July 2025); Bankable Africa (2025); SADV (2020); MAP (2017); South African History Online; Sol Plaatje University; Encyclopaedia Britannica (Lubumbashi); Copperbelt University; University of Mines and Technology; GeoNames gazetteer (regional seats); UNSD (DRC census rounds). n/p: not published in the sources reviewed.

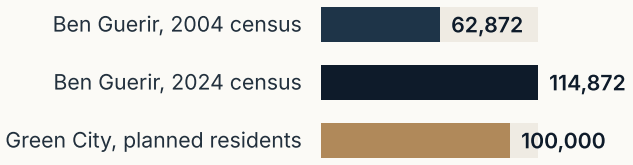
Kimberley is the longest-run example. Mining at the Big Hole stopped in 1914. The city became capital of the Northern Cape when the province was formed in 1994, and Sol Plaatje University opened there in 2014. Lubumbashi, established in 1910 as a copper-mining settlement, has had a university since 1955, according to Encyclopaedia Britannica, and Kitwe has had the Copperbelt University since 1987.

Administrative status has followed mining more recently: Geita is now the seat of its own region, and in 2015 Kolwezi was attached to Lualaba, a new and cobalt-rich province. Diversified cities in our panel grew more slowly than their countries' cities, but none of them shrank.

EXHIBIT 18

OCP is building a second town almost the size of the first

Ben Guerir census population and Green City planned residents



Source: City Population compiling HCP censuses; SADV (OCP Group), Mohammed VI Green City brochure (2020). **Heirstone analysis:** the plan implies 100 residents per gross hectare, 4.0 residents per home and 2 m² of retail space per resident.

A HEIRSTONE FRAMEWORK

Five kinds of mining town, and what moves a town from one to the next

The panel sorts into five types. Each sits at a stage in the life of a mine, and each has a typical growth path. The types are not fixed: Oranjemund moved out of the company-town group by opening its gates, Ben Guerir by adding a university, and Kimberley became a diversified city long after its mine closed. For a government, a miner or an investor, the useful question is which type a town is today and what would move it to the next.

EXHIBIT 19

The Heirstone Mining Town Typology

Five town types with their median growth multiple against the national urban population (1.0 = same pace) in the Heirstone Mining Town Panel

BUILD → OPERATE → MATURE OR CLOSE

TYPE 1 Corridor town

Rail, port and camp towns sized by the construction peak.

EXAMPLES
Simandou rail towns such as Dantilia and Kamara; Morebaya port

MEDIAN MULTIPLE IN THE PANEL
n/a
Not yet measurable

EVIDENCE
8,000 of 10,000 Dantilia workers laid off in three months (Reuters)

WHAT DECIDES THE NEXT STAGE
Whether permanent housing and services are sized for the operating workforce

TYPE 2 Company town

Operator builds the houses, controls residence and is the main employer.

EXAMPLES
Orapa, Jwaneng, OCP and CPG towns, Selebi-Phikwe

MEDIAN MULTIPLE IN THE PANEL
-0.24x



8 towns; 6 shrank

EVIDENCE
Oranjemund (opened) and Ben Guerir (university) broke out

WHAT DECIDES THE NEXT STAGE
Opening land and title to residents and other employers

TYPE 3 Open boomtown

Open land beside an active mine; growth outruns services.

EXAMPLES
Solwezi, Tarkwa, Tete, Geita, Kahama, Letlhakane

MEDIAN MULTIPLE IN THE PANEL
2.16x



6 towns; all faster

EVIDENCE
Adds 12,000 to 21,000 residents a year at peak

WHAT DECIDES THE NEXT STAGE
Serviced land now, and a public function before the boom fades

TYPE 4 Legacy mining city

Long-established town where mining still sets the pace.

EXAMPLES
Kitwe, Chingola, Luanshya, Mufulira, Welkom, Rustenburg, Obuasi

MEDIAN MULTIPLE IN THE PANEL
0.44x



7 towns; none faster

EVIDENCE
Copperbelt towns shrank in the privatisation decade

WHAT DECIDES THE NEXT STAGE
Transferring mine land and housing; a new function or zone

TYPE 5 Diversified city

A capital, university or port carries the town.

EXAMPLES
Kimberley, Gafsa, Lubumbashi

MEDIAN MULTIPLE IN THE PANEL
0.65x



2 towns; none shrank

EVIDENCE
Kimberley a provincial capital 80 years after closure

WHAT DECIDES THE NEXT STAGE
Mining becomes one employer among several

Source: Heirstone Mining Town Panel (Exhibit 3); Reuters via Mining.com (December 2025). **Heirstone analysis.** Corridor towns have no census pair yet; Lubumbashi is a reference without a census pair. Ben Guerir and Oranjemund are counted as company towns that changed model (median 1.56x).

Three readings follow from the typology. Open boomtowns are where demand for housing, retail and services is strongest, but their growth depends on mining investment that has a construction peak and, eventually, an end. Company towns are stable while the mine runs and exposed when it changes, through mechanisation, as in the Gafsa basin, or closure, as at Selebi-Phikwe.

Diversified cities grew at about half to four fifths of their country's urban pace in our panel, and none shrank. The aim of policy is to move a town toward that type while the mine is still paying, which is what OCP is attempting at Ben Guerir and Tanzania at Kahama.

A HEIRSTONE TEST

The city-after-the-mine test: five questions to answer before first ore

The typology describes where a town is. The test below asks whether it is on course to outlive its mine. Each question comes from the evidence in the earlier chapters, and each has a number that a government or an operator could publish. We apply it to six towns using only public information.

EXHIBIT 20

On public information, Ben Guerir and Kahama come closest to passing; the Simandou corridor cannot yet be assessed on four of five tests

Heirstone city-after-the-mine test, applied to six mining towns and corridors, 2026

● Yes ◑ Partly ○ No ? Not answerable from public sources

TEST	QUESTION	EVIDENCE IN THIS REPORT	METRIC TO PUBLISH	SIMANDOU CORRIDOR	KOLWEZI	SOLWEZI	KAHAMA	BEN GUERIR	ORAPA
1. Open land	Can people other than the operator's staff live, buy land and build in the town?	Orapa -9% against Letlhakane +59%; Oranjemund almost doubled after opening	Share of town land titled outside mining leases	?	◑	●	●	◑	○
2. A second function	Does the town have, or have a dated plan for, a function that would keep people if the mine closed?	No diversified city in the panel shrank; Ben Guerir grew 1.76x urban Morocco's pace	Public and institutional jobs as a share of formal jobs	?	●	●	◑	●	○
3. Sized for operations	Are permanent housing and utilities planned for the operating workforce and its induced growth, with the peak in temporary housing?	Simandou peak 4 times operations; 80% of Dantilia workers laid off in three months (Reuters)	Construction-to-operations workforce ratio	?	?	○	?	●	●
4. Revenue that arrives	Is the host share of mining revenue paid and published each year?	37 to 38% of the due share arrived in Katanga and Ghana (2014); Zambia earmarks none	Heirstone delivery ratio	◑	◑	○	?	?	?
5. Closure as handover	Is there a plan to pass land, houses and infrastructure to residents and public bodies before the mine ends?	Buzwagi converted at closure; Selebi-Phikwe replaced about a quarter of jobs after it	Hectares and homes transferred before closure	?	◑	?	●	◑	?

Source: Heirstone assessment from the sources cited in chapters 2 to 6: Debswana; Statistics Botswana; Reuters via Mining.com; EITI; NRGI; GHEITI; Mining Weekly; Bankable Africa; SADV; First Quantum; Times of Zambia (2015); AFP (2022); Radio Okapi (2015). Revenue marks reflect each country's rules: Guinea pools its local fund nationally; the DRC rules exist but payments are partly published; Zambia earmarks none; Tanzanian, Moroccan and Botswanan rules were not verified.

The most common mark in the table is a question mark. For Simandou, the public record cannot answer four of the five questions, because no town or housing plan for the corridor has been published. For Solwezi, the fastest-growing town in the panel, it cannot say what happens at closure. For governments and lenders that gap is the main finding: the information that decides whether a mine leaves a city is rarely published.

Ben Guerir and Kahama come closest to passing, by different routes. Ben Guerir is the corporate route, with the operator as master developer of a university and a second city. Kahama is the public route: an open town where a closed mine is becoming an economic zone run, as proposed, by a national authority. Orapa is the opposite case: well serviced and sized for its workforce, but closed to anyone else.

5

questions, each with a metric a government or operator could publish every year

4 of 5

tests that cannot be assessed for the Simandou corridor from public sources

CHAPTER 7

07 North Africa and Sub-Saharan Africa

The same forces play out differently by country, because land law, revenue rules and the operator's role differ. We start with the Central African Copperbelt, where growth has moved west.

EXHIBIT 21

The Copperbelt's growth has moved west, to Solwezi, Kalumbila and Kolwezi

Mining towns of the Central African Copperbelt; growth multiple against urban Zambia where a census pair exists



Source: Heirstone Mining Town Panel; Zambia Statistics Agency via City Population; Bankable Africa; Natural Earth; town positions from the GeoNames gazetteer. The DRC has held no census in more than 40 years (UNSD). Positions of Kalumbila (at its airport) and Musompo are approximate.

DRC. Kolwezi was attached in 2015 to Lualaba, a new province in the cobalt belt, according to Radio Okapi. Lualaba's provincial revenue directorate collected about US$332m from extractives in 2020-21, 62.8% of the total reported by ITIE-RDC for provincial directorates. From 2025 the province also has the Musompo zone, where works began on more than 900 hectares.

The gaps are data and land. With no census in more than 40 years, any investment case rests on estimates, and the Gécamines quarter shows city districts and mine plans competing for the same ground. A cadastre, an address register and a published boundary between mining titles and urban land would serve the city, its investors and the mines.

Zambia. Solwezi grew 9.9% a year between 2000 and 2022 on the ward-based 2022 count, Kalumbila district reached 177,067 people seven years after it was created, and the four old Copperbelt towns in our panel grew at 0.35 to 0.66 times the urban rate. Gulf capital is now part of the old Copperbelt's story: International Resources Holding, based in Abu Dhabi, completed the purchase of 51% of Mopani for about US$1.1bn in 2024, according to Forbes Middle East. Mopani took over ZCCM's Nkana and Mufulira divisions at privatisation.

None of the US$1.9bn Zambia raised in mining taxes in 2022 is earmarked for host areas, so town finance in Solwezi and Kalumbila depends on national transfers and on the mines. For investors, the North-Western Province combines the country's fastest-growing town with a provincial capital's functions; the old Copperbelt offers larger, slower markets with recapitalised mines.

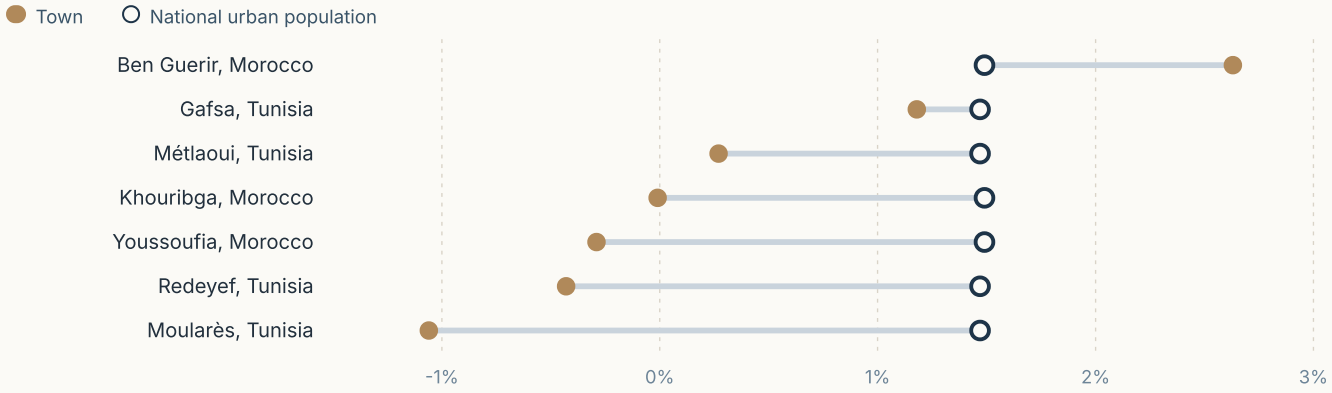
NORTH AFRICA

The phosphate towns of Morocco and Tunisia have levelled off. Ben Guerir is the exception.

EXHIBIT 22

Of seven phosphate towns in Morocco and Tunisia, only Ben Guerir outgrew its country's cities

Annual population growth of the town and of the national urban population, latest intercensal period (Morocco 2014-24, Tunisia 2004-14), %



Source: City Population compiling HCP (Morocco) and INS (Tunisia) censuses; World Bank WDI. **Heirstone analysis.**

Morocco. OCP's three phosphate towns have diverged. Khouribga was flat between 2014 and 2024 and Youssoufia shrank slightly, while Ben Guerir grew 2.6% a year against 1.5% for urban Morocco. All three sit on OCP operations; Ben Guerir is the one where OCP has built a university and a new city. For developers, Ben Guerir tests whether an operator-led knowledge city can sustain a housing and retail market.

Tunisia. In the Gafsa basin, Redeyef and Moularès lost residents between 2004 and 2014 and Métlaoui barely grew, while Gafsa, the governorate capital, grew 1.2% a year. Tunisian commercial phosphate output did not exceed 3.5 Mt a year in 2011-2022, against 8 Mt in 2010 (AGBI), and the Compagnie des Phosphates de Gafsa (CPG) produced 3.03 Mt in 2024. In both countries the town with a role beyond the mine held up best.

Mauritania. SNIM, the state iron-ore company, runs a railway of more than 700 km and the ore-loading port at Nouadhibou, and reported 7,003 direct jobs in 2025. Zouérat and Nouadhibou appear as reference towns on the panel map; comparable town counts for both census rounds were not available to us.

Egypt, Algeria and Libya are outside this edition of the panel, which needs two comparable town-level census counts for each town; their mining districts are candidates for a later edition.

0.0%

Khouribga's annual growth, 2014 to 2024: 196,196 residents, then 195,931

2.6%

Ben Guerir's annual growth over the same decade

3.03 Mt

CPG phosphate output in 2024, against 8 Mt of Tunisian commercial output in 2010

SUB-SAHARAN AFRICA

From Simandou to the Namib, country by country

Beyond the Copperbelt, eight countries show the same pattern at different stages: a corridor still being built in Guinea, company towns in Botswana and Namibia, legacy cities in South Africa and open boomtowns in Ghana and Tanzania.

Guinea

Simandou's corridor runs about 600 km of new railway to the port at Morebaya; its towns have no published population data or town plan in the sources we reviewed. Reuters reports that construction peaked above 60,000 workers, that operations need fewer than 15,000 and that Guinea's Simandou 2040 strategy is costed at about US$200bn.

Priority: A corridor land plan that turns construction camps into serviced plots for operating staff, and published payments of the 0.5% local contribution.

Botswana

Debswana's fenced towns stood still while open Letlhakane grew 59% in 2011-2022. Jwaneng's extension, with about 20 years of mining ahead, leaves time to open land and add functions. At Selebi-Phikwe, replacement jobs reached about a quarter of those lost three years after closure.

Priority: A timetable for opening Orapa, and land for employers beyond the mine at Jwaneng, set while both mines are profitable.

Namibia

Oranjemund almost doubled in 2011-2023 after it was proclaimed and opened, ending about 1,100 residents above the urban trend.

Priority: Plan the opening date and the second function at the start of a mine's life.

South Africa

Rustenburg and Welkom grew at 0.13 and 0.44 times urban South Africa's pace in 2011-2022. Producers are selling houses to employees: Implats 4,400 units, Sibanye-Stillwater 2,083, Harmony 4,175 on offer at 40% off.

Priority: Transfer land, houses and services to households and municipalities before shafts close.

Ghana

Tarkwa, home to the University of Mines and Technology, grew 8.4% a year in 2010-2021. Royalties and mining taxes make up about 40% of the budgets of local governments that receive them, and about 38% of the fund's due share was paid in 2014.

Priority: Publish district-level disbursements each year so lenders can underwrite local services.

Tanzania

Geita and Kahama added about 380,000 people in 2012-2022 and are projected to reach about 0.9 million each by 2050. Buzwagi is the first mine-to-economic-zone conversion in our set.

Priority: Treat both towns as secondary cities for planning and infrastructure finance.

Mozambique

Tete, near the Moatize coal mines, doubled to 305,722 between the 2007 and 2017 censuses.

Priority: Plan for the corridor as well as the pit: rail and port towns share the boom.

Mali

Allied Gold bought 80% of the Sadiola gold mine in western Mali in 2020, with the state holding 20%, according to IAMGOLD. Kayes, capital of the Kayes region, counted 149,129 people in its urban agglomeration in 2009, according to the UN Demographic Yearbook.

Priority: Link mine expansion to services in the regional capital, which already has them.

DATES TO WATCH

Published milestones that will test the arguments in this report, from Kolwezi to Jwaneng.

Mar 2025	DRC	Works begin on the 900 ha Musompo zone near Kolwezi
Nov 2025	Guinea	First Simandou cargo loaded at Morebaya; SimFer ramps up to 60 Mtpa over 30 months
2026	Botswana	NexMetals reports a 72% larger inferred resource at Selebi Main
2030	Tanzania	NBS projects 485,311 people in Geita and 581,548 in Kahama
2034	Botswana	Jwaneng pit expected to reach 816 m under the Cut-8 and Cut-9 extension

WHAT TO DO

Plan the town that will be there after the mine

The decisions that settle whether a mine leaves a city are taken early: where the town's land is, who owns the houses, what the town is for besides the mine, and who pays for it. Each audience holds part of the answer.

FOR

Governments, land agencies and investment promotion agencies

- Zone open, titled land for the town outside the mining lease before the mine is built. Fenced Orapa shrank 9% while open Letlhakane grew 59%.
- Publish a delivery ratio each year: host-level revenue received against the legal entitlement. Where both are published, 37 to 38% arrived.
- Give mining towns a function that does not depend on the mine: a regional office, a university campus or an economic zone, as at Kimberley, Geita, Kahama and Ben Guerir.
- Count the people. The DRC has held no census in more than 40 years; an address register for Kolwezi and Lubumbashi would help every investor there.

FOR

Mining companies

- House the operating workforce permanently and the construction peak temporarily. Simandou's peak was four times its operating need.
- Sell or transfer houses to occupants well before closure, as Implats, Harmony, Sibanye-Stillwater and Assmang are doing.
- Write the closure plan as a land plan that names the public body taking the site and its assets, as at Buzwagi.
- Open company towns on a published timetable. Oranjemund almost doubled after opening.

FOR

DFIs and lenders

- Ask for a town land plan and a construction-to-operations workforce ratio alongside environmental and social studies.
- Finance serviced land and utilities in open boomtowns adding 12,000 to 21,000 people a year; company housing covers a fraction of that.
- Back mine-to-zone conversions that sit with a national zone authority, as proposed for Buzwagi in 2026.

FOR

Developers, investors and corporates

- Favour open towns with a second function and new mine investment nearby: Solwezi, Kahama, Geita, Tarkwa, Ben Guerir.
- Underwrite to the operating workforce: Dantilia lost 80% of its construction jobs in three months (Reuters).
- Price the data gap: no published rent series exists for Kolwezi or Solwezi in the sources we reviewed.
- Site branches and networks by town type: boomtowns for growth, diversified cities for stability, company towns with the operator as anchor client.

How Heirstone works on mining towns and corridors

We help governments, mining companies, development finance institutions and developers plan the town around a mine: population and demand studies, land-use and highest-and-best-use analysis, sizing of housing and infrastructure for operating workforces, feasibility modelling (DCF, IRR, NPV and Monte Carlo testing) for conversions such as economic zones, and benchmarking of the kind in this report.

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NOTES, SOURCES AND DEFINITIONS

Notes

Definitions

Mining town: a town whose founding or growth is tied to a mine, mine-linked processing plant or ore port. **Growth multiple:** the town's compound annual population growth divided by that of its country's urban population over the same census years; 1.0 means the same pace, below zero means the town lost residents. **Company town:** a town built by the operator to house its workforce, where it remains the dominant employer and landlord. **Delivery ratio:** host-level revenue received as a share of the statutory entitlement for the same year. **SEZ:** special economic zone. **MDF:** Ghana's Minerals Development Fund. **FNDL:** Guinea's National Local Development Fund.

Method

The Heirstone Mining Town Panel uses two census counts per town from the same series and World Bank urban population for the same years. Town types are a Heirstone classification based on published descriptions of housing and administrative status. Figures are as published, in the currency and period stated; local currencies are converted at World Bank annual average rates for the year shown (GHS 2.90 in 2014, BWP 10.35 in 2017, R18.45 in 2023, R18.33 in 2024, and for Guinea GNF 8,511 in 2023, the DEC alternative conversion factor, as the official-rate series has no 2023 value) unless the source gives its own conversion. Derived figures are labelled Heirstone analysis and use only the inputs shown. Single-source figures from one press report are attributed in the text. Where sources conflict, we give the range or the more conservative figure. Oranjemund's 2023 count is the town council's census figure; Tete's counts are the preliminary results of both censuses. No DRC town is in the panel, as the country has held no census in more than 40 years. Mine-life end dates were not available for most mines, and no published rent series was found for Kolwezi or Solwezi. The city-after-the-mine marks are a Heirstone assessment on public information as of 2026.

Use of this report

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PANEL AND TOWN GROWTH (CHAPTERS 1, 2, 7)

City Population (Thomas Brinkhoff) compiling census results of the Zambia Statistics Agency, Haut-Commissariat au Plan (Morocco), Institut National de la Statistique (Tunisia) and Ghana Statistical Service; Statistics Botswana, 2022 Population and Housing Census (preliminary and final results) and 2011 census; National Bureau of Statistics Tanzania, Geita and Kahama council population projections; Stats SA, Census 2022 municipal fact sheet and Census 2011 municipal pages; Oranjemund Town Council, town profile and history (2026); Encyclopaedia Britannica (Oranjemund, Tete, Lubumbashi); Mozambique News Reports and Clippings, no. 400 (2018); United Nations Statistics Division, Demographic Yearbook city population (Kayes); UNSD, census dates for the 2020 round (DRC); SNIM, company profile (2025); World Bank, World Development Indicators, SP.URB.TOTL; Debswana, Orapa and Jwaneng mine pages; GeoNames gazetteer for map positions.

HOUSING (CHAPTER 2)

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CONSTRUCTION (CHAPTER 3)

Reuters via Mining.com (18 December 2025); Mining Weekly (18 July 2025); Rio Tinto, Simandou pages; The Maritime Executive (2025); Guinéenews (10 November 2023); First Quantum, Kansanshi and Sentinel pages and Trident presentation (January 2014); Lusaka Times (12 August 2013); Daily Nation, Zambia (6 December 2017); Times of Zambia (2015); Zambia Statistics Agency, 2022 Census Preliminary Report; Lusaka Times (2015).

REVENUE (CHAPTER 4)

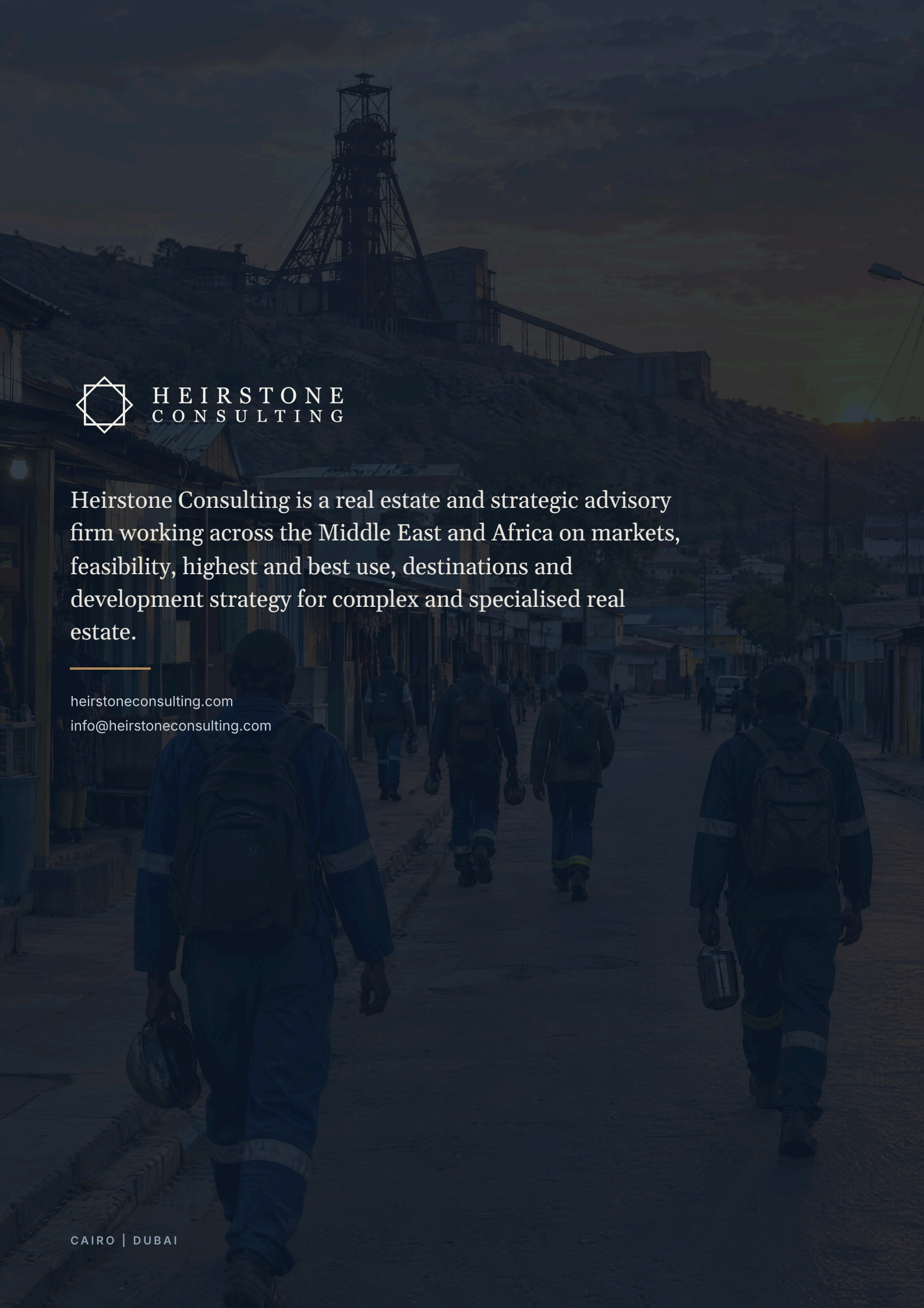
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CLOSURE AND CONVERSION (CHAPTER 5)

IndustriALL (October 2016); Botswana Daily News (26 June 2017, 27 June 2018, 1 October 2019); The Patriot on Sunday (c.2017); NexMetals Mining (2026); Minerals Council South Africa, gold; Mining Weekly (25 July 2025); AllAfrica/Daily News (16 March 2026); Bankable Africa (2025); ZCCM-IH, background and history (2021); Forbes Middle East (2024); News24 (2021); Radio Okapi (2015).

SECOND FUNCTIONS AND NORTH AFRICA (CHAPTERS 6, 7)

SADV (OCP Group), Mohammed VI Green City and development brochures (2020); Bladi.net (5 January 2010); AGBI (2023); La Presse via AllAfrica (23 October 2024); Webdo.tn (9 January 2025); MAP (2017); South African History Online, Northern Cape; Sol Plaatje University; Copperbelt University; University of Mines and Technology; SNIM (2025); Natural Earth; GeoNames.



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